A Constructive Introduction to Quantum Field Theory

Φ^4_3 on the torus I: the FBSDE in force coordinates and renormalisation

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Joint work with Sarah-Jean Meyer, with the use of AI

SISSA lectures, Lecture 6 (provisional) — notes: Lecture 6, §§10–12

Where we are

Lecture 4. Interacting field at resolution t = solution of the FBSDE

$$\Phi_t = \int_0^t \dot{Q}_s F_s \, \mathrm{d}s + \int_0^t \dot{Q}_s^{1/2} \mathrm{d}W_s, \qquad F_t = \mathbb{E}_t\big[-D\,V_T(\Phi_T)\big].$$

In the **forecast coordinate** $\psi_t = \mathbb{E}_t \Phi_T$ the forward equation has no drift and the trees are integrated exactly; the backward half is the **force equation**.

Lectures 6–7. Construct Φ^4_3 on $\mathbb{T}^3$ with this FBSDE:

1. **deterministic renormalisation**: an explicit approximate force $\bar{\mathscr{F}}_q$ and its defect (today);
2. **a priori estimates** for the exact FBSDE in the coordinates of $\bar{\mathscr{F}}_q$, uniform in the lattice (Lecture 7);
3. **removing the cutoffs** and identifying the limit (Lecture 7).

Notes: Lectures 6–7 and the Supplement; Bailleul–Chevyrev–G. (in prep.)

What is proved

For every coupling $\lambda > 0$, every reference mass $q_0 > 0$ and every finite part r of the counterterm, on a fixed torus, with no smallness, and an explicit counterterm with arbitrary finite part:

- the whole lattice sequence converges to a measure ν, an explicit density over an explicit SDE;
- the forecasts converge to the solution of an FBSDE from $\Upsilon_{q_0} = 0$, driven by the limit of the lattice innovations, and $\psi_q = m_q(\Phi_q)$;
- uniqueness in law for its scalar form, a variational formula with no infinite constant, quartic tails, integration by parts.

Not proved: the vector FBSDE alone, strong solutions, OS axioms in full, infinite volume.

Guiding principle. Do not estimate Φ and F. Estimate the error of an explicit approximate forecast.

The results of Lectures 5–7 are new: joint work with S.-J. Meyer + AI.

The problem and the exact FBSDE

Lecture 6

Lattice and the mass-regulator profile

$\Lambda_N = (N^{-1}\mathbb{Z}/\mathbb{Z})^3$, $\int f = N^{-3}\sum_x f(x)$, $\Delta = \Delta_N$ nearest neighbour. Resolution parameter $q = q_0 + t^2 \in [q_0, \infty)$:

$$K_q = q - \Delta, \qquad C_q = K_q^{-1}, \qquad C = C_{q_0} = (m^2 - \Delta)^{-1}.$$

In Lecture 3 notation ($T = \infty$, and $\dot{Q} = \partial_q Q_q$):

$$Q_q = C - C_q, \quad \dot{Q} = C_q^2, \quad \dot{Q}^{1/2} = C_q, \quad S_q = C_q, \quad \mathscr{R}_q = q - q_0.$$

Free resolvents (uniform in N)

$C_q \geqslant 0$ as a kernel, $C_q 1 = q^{-1} 1$, $\|C_q\|_{p\to p} = q^{-1}$,

$$a_2(q) = (C_q^2)(x,x) \lesssim q^{-1/2} \wedge N^3 q^{-2}, \qquad \mathrm{Tr} C \asymp N.$$

The **bubble** $a_2(q)$ is the coincident-point variance of the shell at resolution q.

The cutoff measure

$$U(\varphi)=\tfrac{\lambda}{12}\int\varphi^4,\ V_N=U+\tfrac{\mu_N}{2}\int\varphi^2,\ \nu_N(\mathrm{d}\varphi)=\mathcal{Z}_N^{-1}\mathrm{e}^{-V_N(\varphi)}\mu^C(\mathrm{d}\varphi).$$

So $D^2U(y)=\lambda y^2$: the coupling is the curvature, the same λ as in Lecture 4.

The Wick ordering is now part of μ_N.

Reference mass $q_0>0$ (constants $\in\ker\Delta$ on the torus; r may be negative). Only $q_0+\mu_N$ matters: fix q_0, vary r.

Main result, informal. Let

$$\mu_N=r-\lambda\,\mathrm{Tr}\,C+\frac{2}{3}\lambda^2\int C^3.$$

Then (a) $\nu_N\to\nu$ in H^{-s}, $s>\frac{1}{2}$, along the whole sequence; (b) the forecasts converge in law to a solution of a limiting FBSDE; (c) ν has a variational formula, an integration by parts formula and quartic tails.

These come from bounds on the error Z of an approximate forecast, uniform in N: **UV stability**.

The Polchinski process as a filter

By Lecture 4, in the profile $S_q = C_q$, $\mathscr{R}_q = q - q_0$; at fixed N everything is a theorem. The Polchinski process solves

$$\mathrm{d}\Phi_q = C_q^2 F_q \,\mathrm{d}q + C_q \,\mathrm{d}W_q, \quad F_q = \mathbb{E}_q[-D\,V_N(\Phi_\infty)], \quad \Phi_{q_0} = 0,$$

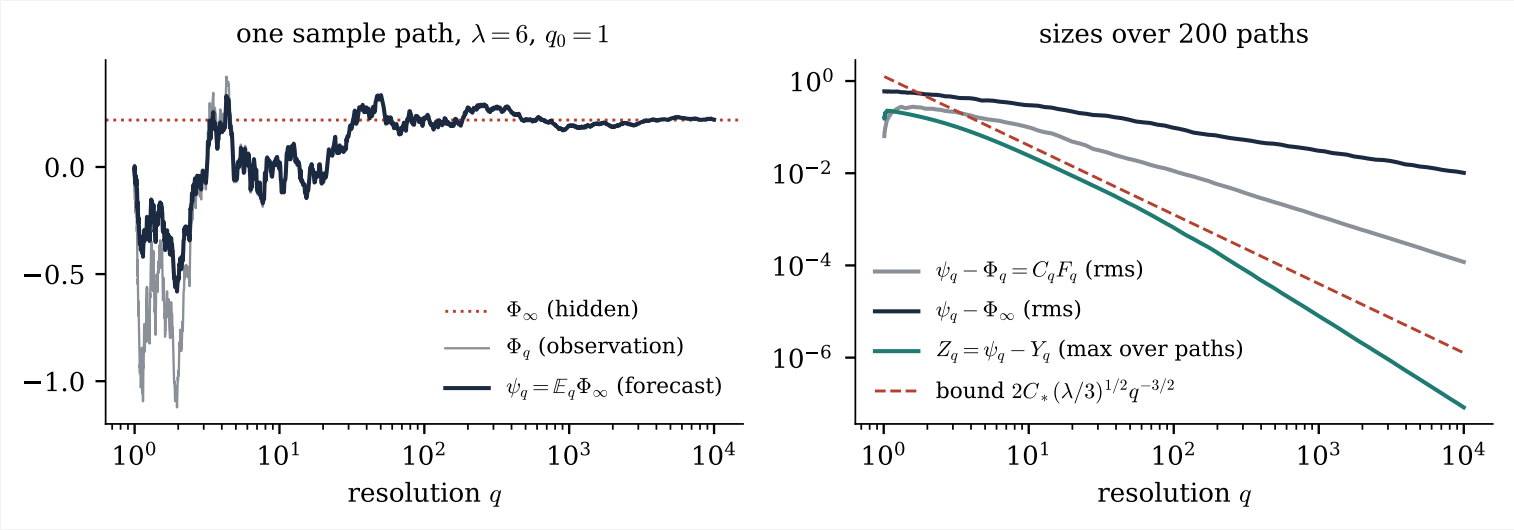
and $\Phi_q \to \Phi_\infty \sim \nu_N$.

- a noisy observation of its endpoint, $\Phi_q = (q - q_0) C_q \Phi_\infty + C_q B_{q-q_0}$, with B independent of Φ_∞;
- forecast

$$\psi_q = \mathbb{E}_q \Phi_\infty = \Phi_q + C_q F_q = m_q(\Phi_q),$$

  covariance $\mathrm{Cov}_q > 0$;
- W is the innovation: $\mathrm{d}\psi_q = \mathrm{Cov}_q \,\mathrm{d}W_q$;
- noise: $\eta_q = K_q \Phi_q - (q - q_0)\psi_q = \mathbb{E}_q B_{q-q_0}$, sub-Gaussian whatever the interaction (L5).

One site



One site, $\nu \propto \mathrm{e}^{-\lambda\varphi^4/12}\mu^{1/q_0}$, $\lambda = 6$, $q_0 = 1$. Left: one path of the observation Φ_q and of the forecast ψ_q, both converging to the hidden Φ_∞. Right, over 200 paths: the forecast error $\sim q^{-1/2}$, the force term $\psi_q - \Phi_q = C_q F_q \sim q^{-1}$, and the error Z_q of the approximate forecast from $\bar{\mathscr{F}} = \frac{\lambda}{3} y^3$, below the bound $2C_*(\lambda/3)^{1/2} q^{-3/2}$ of Theorem 12.8.

The exact force in the forecast coordinate

Since $\Phi_q \mapsto \psi_q$ is invertible, define $\mathscr{F}_q$ by

$$\Phi_q = \psi_q + C_q \mathscr{F}_q(\psi_q)$$

and $H_q^{\mathrm{ex}}(y) = (K_q + D\mathscr{F}_q(y))^{-1}$. It is the $\mathscr{F}$ of Lecture 4 ($x = \psi + S\mathscr{F}(\psi)$), and $F_q = -\mathscr{F}_q(\psi_q)$.

The force equation with a mass regulator

$$\partial_q \mathscr{F}_q + \frac{1}{2}\mathrm{Ctr}_{(H_q^{\mathrm{ex}})^2} D^2 \mathscr{F}_q = 0, \quad \mathrm{Cov}_q = H_q^{\mathrm{ex}}(\psi_q).$$

Itô on $\Phi = \psi + C\mathscr{F}(\psi)$. Its integrated form is Wetterich's equation, which we do not need. An identity about the unknown ν_N, not a construction.

Strategy: an approximate force $\bar{\mathscr{F}}$ gives approximate forecast coordinates; the error should be driven by the failure of $\bar{\mathscr{F}}$ to solve this equation.

Approximate forecast and error

$\bar{\mathscr{F}}_q$ odd, $D\bar{\mathscr{F}}_q$ symmetric, $K_q + D\bar{\mathscr{F}}_q \geqslant \bar{c}_q$. Recall $\psi_q = \mathbb{E}_q[\Phi_T]$, $\mathrm{Cov}_q = \mathrm{Var}_q(\Phi_T)$

$$H_q = (K_q + D\bar{\mathscr{F}}_q)^{-1}, \quad \mathscr{D}_q = \partial_q \bar{\mathscr{F}}_q + \frac{1}{2}\mathrm{C}\mathrm{tr}_{H_q^2} D^2 \bar{\mathscr{F}}_q, \quad b_q = -H_q \mathscr{D}_q.$$

$$\Upsilon_q + C_q \bar{\mathscr{F}}_q(\Upsilon_q) = \Phi_q, \quad Z_q = \psi_q - \Upsilon_q, \quad \Xi_q = \mathrm{Cov}_q - H_q(\Upsilon_q).$$

Exact, at fixed N

$$\mathrm{d}\Upsilon_q = [b_q(\Upsilon_q) + H_q(\Upsilon_q) Z_q]\,\mathrm{d}q + H_q(\Upsilon_q)\,\mathrm{d}W_q, \qquad \Upsilon_{q_0} = 0$$

$$\mathrm{d}Z_q = -[b_q(\Upsilon_q) + H_q(\Upsilon_q) Z_q]\,\mathrm{d}q + \Xi_q\,\mathrm{d}W_q, \quad Z_\infty = \Phi_\infty - \Upsilon_\infty.$$

Proof: Itô on the coordinate equation, with the Polchinski process and $\mathrm{d}C_q = -C_q^2 \mathrm{d}q$: the martingale part gives H, the drift gives $HZ + b$.

Reading the system

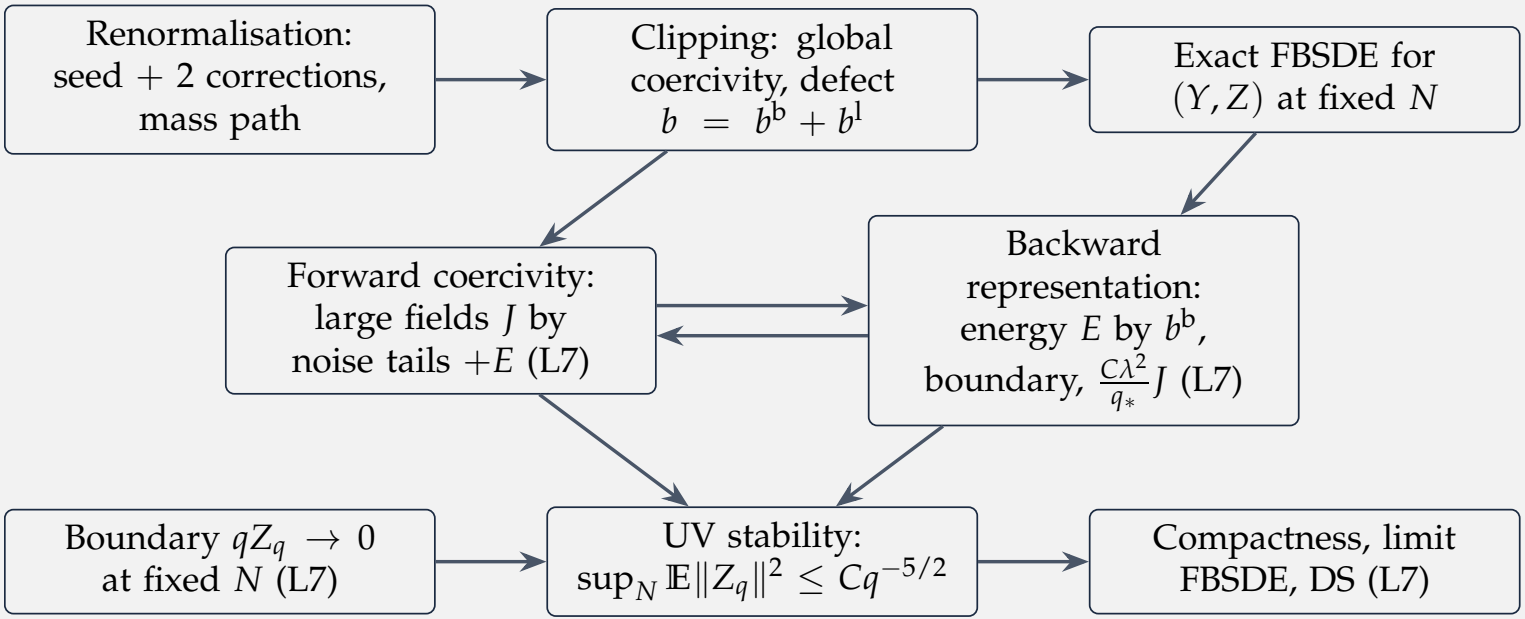
- Y: forward and explicit once Z is known; coefficients b, H built from $\bar{\mathscr{F}}$.
- Z: backward. The terminal condition $q\,Z_q \to 0$ is **proved** at fixed N (Lecture 7), because $\bar{\mathscr{F}}_q \to$ bare force.
- The only source of Z is the drift b, the force defect, transported by $\exp \int H$.

UV stability

The explicit force $\bar{\mathscr{F}}$ computes the forecast up to an error Z which is small uniformly in N.

Needed: (i) a defect integrable against the backward propagator (**renormalisation**); (ii) global coercivity of $\bar{\mathscr{F}}$ (**large fields**); (iii) a priori estimates (Lecture 7).

The structure of the construction



Renormalisation: why two of them

Lecture 6

Approximating the Wetterich force

We seek an explicit force with a small **defect**, not the full solution of the nonlinear flow.

$$\partial_q \mathscr{F}_q = -\frac{1}{2} C \mathrm{tr}_{H_q^2} D^2 \mathscr{F}_q, \qquad H_q = (K_q + D\mathscr{F}_q)^{-1}.$$

At infinite resolution: $\mathscr{F}_\infty = U' + \mu_N y$, with $U'(y) = \frac{\lambda}{3} y^3$.

1. Insert the bare cube into the response and the vertex: a **one-loop seed**.
2. Feed each approximation back into the equation: **two correction steps**.
3. Estimate the remaining defect in the backward equation for Z.

First separate the linear flow: it produces the divergent mass counterterm and cannot be treated as a small nonlinear error.

Why the linear term must be separated

Insert U' into the loop, with $G_q(y) = (K_q + \lambda y^2)^{-1}$:

$$-\frac{1}{2}\mathrm{Ctr}_{G_q^2} D^2 U'(y)(x) = -\lambda y(x) G_q(y)^2(x,x).$$

Its linear Taylor coefficient is

$$-\lambda a_2(q) y(x), \qquad a_2(q) = C_q^2(x,x).$$

In three dimensions $a_2(q)$ has continuum size $q^{-1/2}$: its integral to infinity diverges. At fixed lattice cutoff it is finite, but grows with N.

This is the **tadpole**. Further insertions also produce the logarithmic **sunset**. Keep the linear flow in a running mass; iterate only the nonlinear remainder.

The projection Π

For a smooth odd force f, define

$$(\Pi f)(y) = f(y) - \mathrm{D}f(0)y, \qquad ((1-\Pi)f)(y) = \mathrm{D}f(0)y.$$

- $\Pi^2 = \Pi$ and $D(\Pi f)(0) = 0$: the remainder starts at **cubic order**.
- We remove the **whole linear operator**, including its nonlocal part. The cubic interaction and higher vertices remain.

$$\mathscr{F}_q = U' + M_q y + \ell_q, \qquad \Pi \ell_q = \ell_q, \quad M_q = \mu_q I + \Theta_q.$$

Nonlinear part: integrate from ∞ with $\ell_\infty = 0$. Linear part: solve the mass flow; fix the scalar mass at q_* and the centred kernel at infinity.

Scales and power counting

$$g_q = \lambda q^{-1/2} (\text{dimensionless coupling}),$$

$$r_q = (q/\lambda)^{1/2} (\lambda r_q^2 = q : \text{nonlinearity} \approx \text{regulator}).$$

Free field at resolution q: $\mathrm{Var} X_q(x) \lesssim q^{1/2}$, typical size $q^{1/4} = g_q^{1/2} r_q \ll r_q$.

Rooted norm of a vertex $T(x_1, \dots, x_k)$ and field ball:

$$s(T) = \max_i \sup_{x_i} \int |T| \prod_{j \neq i} \mathrm{d}x_j, \qquad \mathsf{S}_k(g) = \sup_{\mathcal{B}_q} s\,(D^{k-1} g), \qquad \mathcal{B}_q = \{\|y\|_\infty < 2 r_q\}.$$

Natural size of the kth vertex of the **loop corrections**:

$$\mathsf{S}_k \lesssim g_q \, q \, r_q^{2-k} \qquad (k = 4 : \lambda \, g_q).$$

The loop estimate

Loop estimate

A closed loop with p vertices: $s(\text{loop}) \leqslant C a_2(q)\, q^{-(p-1)} \prod_i s(T_i)$ (two propagators in L^2, the others in operator norm). After the q integration a one-loop term is of size $g_q q r_q^{2-k}$; every further loop gains a factor g_q.

Computing the one-loop seed

Project the bare loop before integrating:

$$\Pi[-\lambda y(x) G_q(y)^2(x,x)] = -\lambda y(x)[G_q(y)^2(x,x) - a_2(q)].$$

Use $\partial_q G_q = -G_q^2$ and $\partial_q C_q = -C_q^2$. With zero boundary value at infinity:

$$\ell_q(y)(x) = \lambda y(x)[G_q(y)(x,x) - C_q(x,x)].$$

The leading term is the bubble:

$$-\lambda^2 y(x) \int C_q(x,z)^2 y(z)^2 \mathrm{d}z.$$

Four-leg size: $S_4(\ell_q) \lesssim \lambda^2 q^{-1/2} = \lambda g_q$. This is level 0: $\ell^{[0]} = \ell$.

Splitting the flow and deriving the iteration

For $\mathscr{F} = U' + My + \ell$, define the loop map

$$\mathscr{T}_q[M, \ell] = -\frac{1}{2}\mathrm{C}\mathrm{tr}_{R^2} D^2(U' + \ell), \quad R = (K_q + M_q + D(U' + \ell))^{-1}.$$

The projections split the force equation:

$$\partial_q \ell_q = \Pi \mathscr{T}_q[M, \ell], \qquad \partial_q M_q = D\mathscr{T}_q[M, \ell](0).$$

Hold the mass path fixed for the nonlinear iteration. Integrate the projected flow from infinity and insert the preceding approximation:

$$\ell^{[0]} = \ell, \qquad \ell_q^{[j+1]} = -\int_q^\infty \Pi \mathscr{T}_s[M, \ell^{[j]}]\mathrm{d}s, \quad j = 0, 1.$$

The seed satisfies $\partial_q \ell_q = \Pi \mathscr{T}_q[0, 0]$. The output is $\ell^{[2]}[M]$: a function of the trial mass path, which is determined by a separate fixed point.

First correction: insert the seed

Set $R^{[j]} = (K + M + \mathrm{D}U' + D\ell^{[j]})^{-1}$ and $\delta^{[1]} = \ell^{[1]} - \ell$. Subtract the bare loop from the loop containing the seed:

$$\delta_q^{[1]} = \frac{1}{2}\int_q^\infty \Pi[\mathrm{Ctr}_{(R^{[0]})^2 - G^2} D^2 U' + \mathrm{Ctr}_{(R^{[0]})^2} D^2 \ell\,]_s \mathrm{d}s.$$

Two changes: the propagators and the vertex.

$$R^{[0]} - G = -R^{[0]}(M + D\ell)G.$$

The insertion $(M + D\ell)/q$ is $O(g_q)$. One extra loop gives

$$\mathsf{S}_4(\delta_q^{[1]}) \lesssim \int_q^\infty g_s s^{1/2} r_s^{-4} \mathrm{d}s = \lambda^3 q^{-1} = \lambda g_q^2.$$

All factors inside the integral are evaluated at scale s.

Second correction: insert the first correction

Put $\delta^{[2]} = \ell^{[2]} - \ell^{[1]}$. Subtract the first two Picard equations:

$$\delta_q^{[2]} = \frac{1}{2}\int_q^\infty \Pi[\mathrm{Ctr}_{(R^{[1]})^2-(R^{[0]})^2} D^2(U'+\ell) + \mathrm{Ctr}_{(R^{[1]})^2} D^2\delta^{[1]}]_s \mathrm{d}s.$$

The marked insertion is now the first correction itself:

$$R^{[1]} - R^{[0]} = -R^{[1]} D\delta^{[1]} R^{[0]}.$$

$$\mathsf{S}_4(\delta_q^{[2]}) \lesssim \int_q^\infty g_s^2 s^{1/2} r_s^{-4} \mathrm{d}s = \frac{2}{3}\lambda^4 q^{-3/2}.$$

Four-leg sizes: seed λg_q, first correction λg_q^2, second correction λg_q^3.

For squared propagators use $A^2 - B^2 = A(A-B) + (A-B)B$; no commutativity is assumed.

Following a spatial distance through a loop

At zero field, set $W_q(T) = s(T) + \sqrt{q}\, v_1(T)$. Mark two external legs and follow their distance around the loop.

- A displacement inside a vertex is controlled by that vertex's first moment.
- A displacement along an inverse edge is controlled by its weighted L^1 or L^2 bound. The triangle inequality distributes the external distance over these pieces.

For p vertices and $p+1$ inverse factors:

$$W_q(\mathscr{Z}) \lesssim q^{-1/2} q^{-(p-1)} \prod_{i=1}^{p} W_q(T_i).$$

The weight $\sqrt{q}$ makes a distance of order $q^{-1/2}$ dimensionless: the weighted loop has the same scale powers as the unweighted loop.

Keep within-vertex moments: smoothing the propagators cannot supply spatial localisation for an arbitrary nonlocal vertex.

Integrating preserves the moment weight

A loop difference with relative size ε_s has a marked vertex or inverse. The weighted comparison gives

$$W_s(\mathrm{D}^{k-1}\delta\mathscr{T}_s(0)) \lesssim \varepsilon_s s^{1/2} r_s^{-k}.$$

For $s \geqslant q$, $\sqrt{q} \leqslant \sqrt{s}$:

$$W_q\Big(\int_q^\infty A_s \mathrm{d}s\Big) \leqslant \int_q^\infty (s(A_s) + \sqrt{q} v_1(A_s))\mathrm{d}s \leqslant \int_q^\infty W_s(A_s)\mathrm{d}s.$$

First correction: $\varepsilon_s \lesssim g_s$. Second correction: $\varepsilon_s \lesssim g_s^2$. The same gains hold for the spatial moment and for the rooted norm.

Π subtracts only a linear force, so it leaves all $k \geqslant 4$ vertices unchanged. The field supremum also propagates, since $\mathscr{B}_q \subseteq \mathscr{B}_s$.

The four-leg moment after two corrections

Write $T_q^{[j]} = \mathrm{D}^3 \ell_q^{[j]}(0)$. The seed is the bubble:

$$W_q(T_q^{[0]}) \lesssim \lambda^2 \int (1 + \sqrt{q} d(0,z)) C_q(z)^2 \mathrm{d}z \lesssim \lambda^2 q^{-1/2}.$$

For the increments, integrate the weighted loop bounds:

$$W_q(T_q^{[1]} - T_q^{[0]}) \lesssim \lambda^3 \int_q^\infty s^{-2} \mathrm{d}s = \lambda^3 q^{-1},$$

$$W_q(T_q^{[2]} - T_q^{[1]}) \lesssim \lambda^4 \int_q^\infty s^{-5/2} \mathrm{d}s = \frac{2}{3} \lambda^4 q^{-3/2}.$$

With $T_q = T_q^{[2]}$, $s(T_q) + \sqrt{q} v_1(T_q) \lesssim \lambda^2 q^{-1/2}(1 + g_q + g_q^2) \lesssim \lambda^2 q^{-1/2}$.

The $10 \to 8 \to 6$ leg budget supplies the higher vertices needed by both comparisons. Thus $v_1(T_q) \lesssim \lambda^2 q^{-1}$.

The defect after the second correction

The bounded-field approximation is

$$\mathscr{F}_q^o = U' + M_q y + \ell_q^{[2]}.$$

At the mass fixed point: $\partial_q M_q = D\mathscr{T}_q[M, \ell^{[2]}](0)$. The nonlinear iterate uses the preceding level:

$$\partial_q \ell_q^{[2]} = \Pi \mathscr{T}_q[M, \ell^{[1]}].$$

Hence the defect is the loop difference of the last two levels:

$$\mathscr{D}_q^o = \Pi(\mathscr{T}_q[M, \ell^{[1]}] - \mathscr{T}_q[M, \ell^{[2]}]).$$

One loop with an insertion of $\delta^{[2]}$, **without a scale integration**:

$$\mathsf{S}_4(\mathscr{D}_q^o) \lesssim g_q^3 q^{1/2} r_q^{-4} = \lambda^5 q^{-3}.$$

Why stop at the second level?

On the field ball the defect has no linear term, so

$$|b_s| \lesssim s^{-1} \mathsf{S}_4(\mathscr{D}_s) r_s^3, \qquad Z_q \approx \int_q^\infty \frac{s}{q} b_s \mathrm{d}s.$$

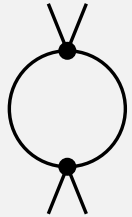
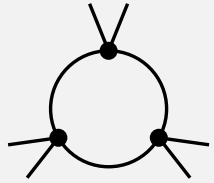
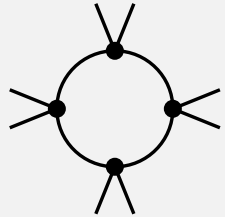
The factor s/q is the backward transport. The source estimates are:

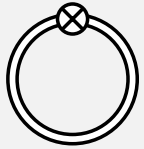
corrections	$\mathsf{S}_4(\mathscr{D}_s)$	backward source integral
0: seed	$\lambda^3 s^{-2}$	$\lambda^{3/2} q^{-1} \int_q^\infty s^{-1/2} \mathrm{d}s = \infty$
1	$\lambda^4 s^{-5/2}$	$\lambda^{5/2} q^{-1} \int_q^\infty s^{-1} \mathrm{d}s = \infty$
2	$\lambda^5 s^{-3}$	$\lambda^{7/2} q^{-1} \int_q^\infty s^{-3/2} \mathrm{d}s = 2\lambda^{7/2} q^{-3/2}$

One correction is logarithmically borderline. **Two corrections beyond the seed suffice** for the bounded-field source estimate.

Each step costs two field derivatives: keep the iteration finite. Full estimates: Lecture 7.

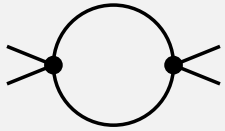
Seed and corrections in graphs

$$\ell_q(y) \;=\; D\,\Big[-\tfrac{1}{4}$$  $$+\tfrac{1}{6}$$  $$-\tfrac{1}{8}$$  $$+\;\cdots\;\Big]$$

$$\partial_q \ell_q^{[j+1]} \;=\; -\Pi\;\tfrac{1}{2}$$  $$\ell[j]$$

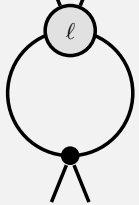
double line: $(K_q + M_q + D(U' + \ell^{[j]}))^{-1}$
cross: $\partial_q \mathcal{R} = 1$

seed $\ell^{[0]} = \ell$
one loop, g_q



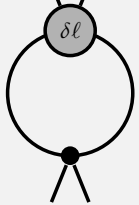
$\longrightarrow$

$\ell^{[1]} - \ell^{[0]}$
two loops, g_q^2



$\longrightarrow$

$\ell^{[2]} - \ell^{[1]}$
three loops, g_q^3



$\Longrightarrow$

stop: defect
$\mathcal{D}^{\mathrm{o}} = \mathcal{O}(g_q^3)$
integrable source

Each step inserts the previous correction into the loop: one more loop, one more g_q. Two steps give a defect of order g_q^3.

The mass

Impose the linear flow for the final approximation, with $\ell^{[2]} = \ell^{[2]}[M]$:

$$\partial_q M_q = -\lambda \mathrm{Tr}(R_{0,q}^2) I - \frac{1}{2} C \mathrm{tr}_{R_{0,q}^2} D^3 \ell_q^{[2]}(0), \quad R_{0,q} = (K_q + M_q)^{-1},$$

with $\mu_{q_*} = b$, $|b| \leqslant B\lambda q_*^{1/2}$, at an **auxiliary scale** q_*, and $\Theta_q \to 0$ as $q \to \infty$. Bare mass $\mu_N = \lim_{q\to\infty} \mu_q$: an output of the flow.

For q_* large, uniformly in N. Unique solution in a ball, $|\mu_q| \leqslant C\lambda q^{1/2}$, $v_1(\Theta_q) \leqslant C\lambda^2 q^{-1/2}$; increments $\mathcal{N}_k(\ell^{[j]} - \ell^{[j-1]}) \leqslant C_k g_q^{j+1} q\, r_q^{2-k}$; and the defect of $\mathscr{F}^\circ$: $S_4(\mathscr{D}_q^\circ) \leqslant C\lambda^5 q^{-3}$.

Explicit counterterm. $\mu_N = r - \lambda \mathrm{Tr}\, C + \frac{2}{3}\lambda^2 \int C^3$ is admissible: the measure does not depend on q_*.

Closing the mass path

Given a trial path M, compute the two corrections, then define

$$B_q[M] = D\mathscr{T}_q[M, \ell^{[2]}[M]](0), \quad B_q[M]1 = \beta_q[M]1.$$

Integrate with the mixed boundary conditions to obtain a new path:

$$\mu_q^{\text{new}} = b + \int_{q_*}^{q} \beta_s[M]\,\mathrm{d}s,$$

$$\Theta_q^{\text{new}} = -\int_{q}^{\infty} (B_s[M] - \beta_s[M]I)\,\mathrm{d}s.$$

$\mathscr{A}[M] := \mu^{\text{new}} I + \Theta^{\text{new}}$. The mass path is the **fixed point** $M = \mathscr{A}[M]$.

This is an equation for the whole path: the nonlinear correction at q depends on M at later scales.

Centring the source removes local terms

Separate the mass source into the tadpole and the correction vertex:

$$B_q = -\lambda \mathrm{Tr}(R_{0,q}^2) I + E_q, \quad E_q = -\frac{1}{2} \mathrm{C} \mathrm{tr}_{R_{0,q}^2} T_q, \quad T_q = D^3 \ell_q^{[2]}(0).$$

Write $E_q 1 = e_q 1$. The scalar source contains the whole local contribution:

$$\beta_q = -\lambda \mathrm{Tr}(R_{0,q}^2) + e_q, \qquad B_q - \beta_q I = E_q - e_q I.$$

The tadpole cancels; so does any term $c_q I$. Off-diagonal entries stay unchanged. Since $d(0,0) = 0$,

$$v_1(B_q - \beta_q I) = v_1(E_q).$$

Centring removes the local obstruction. The remaining spatial estimate must come from the first moment of the nonlinear vertex.

The mass flow produces the first-moment bound

The weighted four-leg estimate gives

$$s(T_q) + \sqrt{q}\, v_1(T_q) \lesssim \lambda^2 q^{-1/2}, \quad v_1(T_q) \lesssim \lambda^2 q^{-1}.$$

Keep the distance between the two uncontracted legs inside T_q. Since $\sup_{u,v} |R^2_{0,q}(u,v)| \lesssim q^{-1/2}$,

$$v_1(B_q - \beta_q I) = v_1(E_q) \lesssim q^{-1/2} v_1(T_q) \lesssim \lambda^2 q^{-3/2}.$$

Integrate the centred flow with zero boundary value at infinity:

$$v_1(\Theta_q^{\mathrm{new}}) \leqslant \int_q^\infty v_1(B_s - \beta_s I)\,\mathrm{d}s \lesssim \lambda^2 \int_q^\infty s^{-3/2}\,\mathrm{d}s = 2\lambda^2 q^{-1/2}.$$

The update preserves the first-moment bound, uniformly in N. At the mass fixed point this is the bound on Θ_q, and it tends to zero as $q \to \infty$.

Two distinct iterations

Inner recursion: for each trial M, start from the seed and compute exactly two nonlinear corrections.

$$M \longmapsto \ell^{[1]}[M] \longmapsto \ell^{[2]}[M] \longmapsto B[M] \longmapsto \mathscr{A}[M].$$

Outer contraction: start at $M_q^{(0)} = b\,I$; iterate $M^{(n+1)} = \mathscr{A}[M^{(n)}]$.

$$\|\mathscr{A}[M] - \mathscr{A}[\hat{M}]\|_* \leqslant C\,g_{q_*}\|M - \hat{M}\|_*.$$

The path norm weights μ_q by $\lambda\sqrt{q}$ and $v_1(\Theta_q)$ by $\lambda^2 q^{-1/2}$. For large q_* the map preserves the ball and contracts, uniformly in N.

Every outer step recomputes the same two-level force. Its depth stays 2, so no additional field derivatives are consumed. At the fixed point the linear flow is exact; the nonlinear defect remains.

Centre before taking absolute values

For an even convolution kernel, $\Theta 1 = 0$ means that the diagonal cancels the off-diagonal mass. With $\delta_z^2 = \sigma_z + \sigma_{-z} - 2I$,

$$C_q\Theta = \frac{1}{2}\int \Theta(z)\,\delta_z^2 C_q \mathrm{d}z.$$

The absolute row norm discards the cancellation. For a lattice step h in direction e, take

$$\Theta_h = \frac{a}{2h}(\sigma_{\mathrm{he}} + \sigma_{-\mathrm{he}} - 2I), \qquad v_1(\Theta_h) = a, \quad \|\Theta_h\|_{\mathrm{row}} = \frac{2a}{h}.$$

The row norm diverges as $h \to 0$; the first moment stays bounded.

Estimate the signed product $C_q\Theta$ first. The quantity needed by the loop is the kernel after resolvent smoothing, not the absolute mass of Θ itself.

Smoothing turns cancellation into a moment bound

The resolvent smooths on length scale $q^{-1/2}$:

$$\|\delta_z^2 C_q\|_{\mathrm{row}} \leqslant C q^{-1/2} d(0,z).$$

Scale check: integrate the heat-kernel translation bound $C d(0,z) t^{-1/2}$ against $\mathrm{e}^{-\mathrm{qt}}\mathrm{d}t$.

$$\|C_q \Theta\|_{\mathrm{row}} \leqslant C q^{-1/2} \int d(0,z) |\Theta(z)| \mathrm{d}z = C q^{-1/2} v_1(\Theta).$$

The path estimate supplies exactly the compensating moment:

$$v_1(\Theta_q) \lesssim \lambda^2 q^{-1/2} \quad \Longrightarrow \quad q^{-1/2} v_1(\Theta_q) \lesssim \lambda^2 q^{-1} = g_q^2.$$

A pure squared-distance bound has a logarithmic loss at kinetic order two. One distance power, with $q^{-1/2}$, suffices. The weighted loop estimate is Lemma 19.3 of the Supplement.

The small factor in a resolvent comparison

The field-dependent response also receives this cancellation. With $V = \lambda y^2$ and $G = (K_q + V)^{-1}$,

$$G\Theta = (I - \mathrm{GV})C_q\Theta, \qquad \|I - \mathrm{GV}\|_{\infty\to\infty} \leqslant 2.$$

The bounded Neumann series for the full R then gives

$$\|R\delta M\|_{\infty\to\infty} \leqslant C\rho_q, \quad \rho_q = \frac{|\delta\mu_q|}{q} + q^{-1/2}v_1(\delta\Theta_q).$$

Holding the nonlinear correction fixed, $\delta R = -R\delta M\hat{R}$ has row norm $\lesssim \rho_q/q$ and row L^2 norm $\lesssim \rho_q\sqrt{a_2(q)}$.

A marked propagator gains the relative factor $\rho_q \lesssim (g_q + g_q^2)\|\delta M\|_*$: g_q from the scalar mass, g_q^2 from the centred kernel.

Mass differences gain a small insertion

Compare two paths with the same b. Write $d = \|M - \hat{M}\|_*$ and δ for their output differences.

$$R^{[j]} - \hat{R}^{[j]} = -R^{[j]}(\delta M + D\delta\ell^{[j]})\hat{R}^{[j]}.$$

The relative scalar and centred insertions obey

$$\frac{|\delta\mu_q|}{q} \leqslant g_q d, \qquad q^{-1/2} v_1(\delta\Theta_q) \leqslant g_q^2 d.$$

Loop comparisons through the same two levels give $\mathcal{N}_{k,q}(\delta\ell^{[j]}) \lesssim g_q^2 q\, r_q^{2-k} d$, $j = 1, 2$, in the available derivative ranges.

The resulting mass-source differences are

$$|\delta\beta_q| \leqslant C\lambda^2 q^{-1} d, \qquad v_1(\delta B_q - \delta\beta_q I) \leqslant C\lambda^3 q^{-2} d.$$

The centred estimate keeps the spatial moment in the nonlocal vertex; the local quartic contributes only to the scalar.

Integration leaves the factor g_{q_*}

Scalar update: the common boundary value b cancels. With $x = q/q_* \geqslant 1$,

$$|\delta\mu_q^{\text{new}}| \leqslant C\lambda^2 d \int_{q_*}^{q} \frac{\mathrm{d}s}{s} = C\lambda^2 \log x\, d,$$

$$\frac{|\delta\mu_q^{\text{new}}|}{\lambda\sqrt{q}} \leqslant C g_{q_*} \frac{\log x}{\sqrt{x}} d \leqslant \frac{2C}{\mathrm{e}} g_{q_*} d.$$

Centred update: integrate from infinity, then divide by its weight.

$$\frac{v_1(\delta\Theta_q^{\text{new}})}{\lambda^2 q^{-1/2}} \leqslant \frac{C\lambda^3 d \int_q^{\infty} s^{-2}\mathrm{d}s}{\lambda^2 q^{-1/2}} = C g_q d \leqslant C g_{q_*} d.$$

Taking both suprema gives the contraction constant $C g_{q_*}$. Fix an invariant ball, then choose q_* so that $C g_{q_*} < 1/2$, uniformly in N.

Why the derivative budget stays fixed

A loop needs $k+2$ legs to control a k-leg vertex. Every evaluation of $\mathscr{A}$ uses the same budget:

$$\text{seed: } 10 \longrightarrow \text{level 1: } 8 \longrightarrow \text{level 2: } 6 \longrightarrow \text{defect: } 4.$$

A k-leg force vertex has $k-1$ field derivatives. The mass source uses only the four-leg vertex at level 2.

- A path difference inserts δM in a resolvent. It is field-independent: $D_y \delta M = 0$. The comparison uses the same field-derivative ranges as the construction.
- $\mathscr{A}$ returns a **mass path**. At the next outer step, restart from the original seed and perform the same two steps. We do not use the previous level 2 as a new seed.

The outer iteration contracts in one fixed space of mass paths. The finite Lipschitz estimates transfer convergence to the corrections; their derivative budget does not shrink with the outer index.

Two mass renormalisations

The two divergent self-energies of Appendix A.2 of Lecture 5:

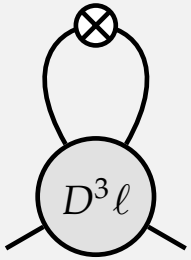
- **Tadpole**: $-\lambda \int_{q_*}^{\infty} \mathrm{Tr} R_{0,q}^2 \, \mathrm{d}q \approx -\lambda \mathrm{Tr} C \sim -N$. Wick ordering: $\frac{\lambda}{12}\varphi^4 - \frac{1}{2}\lambda c \varphi^2 = \frac{\lambda}{12} H_4(\varphi; c) + \text{const.}$
- **Sunset**: contracting the bubble in $D^3\ell(0)$ with R_0^2 and integrating gives the two-loop self-energy,

$$\frac{2}{3}\lambda^2 \int C^3 \approx \frac{\lambda^2}{24\pi^2} \log N \quad (=2\gamma_T \text{ of Lecture 5}).$$

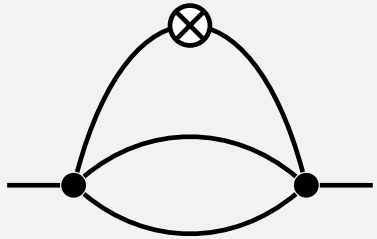
Power counting: nothing else. In the flow: the integrals from $q = \infty$ of all vertices with $k \geqslant 4$ legs converge; only the linear jet needs a boundary condition.

μ_q at fixed q has a limit as $N \to \infty$; μ_N does not. **Never interchange the two limits.**

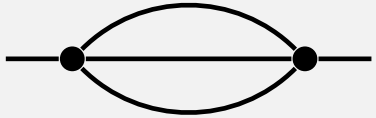
The two mass renormalisations in graphs

$$\partial_q M_q \;=\; -\tfrac{1}{2}$$  $$-\;\tfrac{1}{2}$$ 

tadpole, $D^4U = 2\lambda$

$$\ni$$ 

sunset

$$\mu_N \;\approx\; -$$  $$+$$ 

$\lambda \operatorname{Tr} C \sim \lambda N$ $\qquad$ $\frac{2}{3}\lambda^2 \int C^3 \sim \lambda^2 \log N$

the q-integral removes
the cross: $\int C_q^2\, dq$ closes a line

The cross is $R^2_{0,q}$. The bubble in $D^3\ell^{[2]}(0)$ produces the sunset.

The flow at large fields

Lecture 6

The large-field problem

All bounds hold on $\mathscr{B}_q$ only. The free field exceeds r_q with small, nonzero probability, and nothing a priori controls the interacting forecast. We need $\bar{\mathscr{F}}_q$ defined and uniformly coercive for all fields.

Scalar model

$$\ell_q(y) = \lambda y\big[(q+\lambda y^2)^{-1} - q^{-1}\big]$$

: power series of radius exactly r_q (singularities at $\pm \mathrm{i} r_q$). Half of the truncations destroy monotonicity. The exact function is harmless, $|\ell_q'| \leqslant \frac{9}{8}\lambda/q$.

Perturbative control for $|y| \lesssim r_q$; outside, a non-perturbative replacement that keeps the structure the estimates use: coercivity, carried by the bare cube.

One site: $qy + \frac{\lambda}{3}y^3 = q\,\Phi_q$, noise coefficient $H_q(y) = (q+\lambda y^2)^{-1} \leqslant q^{-1} \wedge (\lambda y^2)^{-1}$. Large fields are more elliptic and less noisy.

Clipping

τ smooth odd, $\tau(u)=u$ on $[-1,1]$, constant outside $[-2,2]$, $0\leqslant\tau'\leqslant 1$;

$$\kappa_q(y)=r_q\tau(y/r_q),\quad \tilde{\ell}_q=D\kappa_q\cdot\ell_q^{[2]}\circ\kappa_q,\quad \bar{\mathscr{F}}_q=U'+M_qy+\tilde{\ell}_q.$$

The cubic and linear parts are not clipped, and $D\kappa_q$ keeps the force a gradient. Globally: $s(D^{j-1}\tilde{\ell}_q)\leqslant C g_q q\, r_q^{2-j}$ $(j=2,3,4)$, $s(D\partial_q\tilde{\ell}_q)\leqslant C g_q$.

Global coercivity

For all y: $c(K_q+\lambda y^2)\leqslant H_q(y)^{-1}\leqslant C(K_q+\lambda y^2)$,

$$\|H_q(y)\|_{p\to p}\leqslant\frac{1+C(g_q+g_q^2)}{q},\quad \operatorname{diag}H_q(y)^2\leqslant C a_2(q).$$

Neumann series for $H=(I+G(M+D\tilde{\ell}))^{-1}G$, $G=(K_q+\lambda y^2)^{-1}\leqslant C_q$.

The complete defect

$b = b^{\mathsf{b}} + b^{\mathsf{I}}$, $b^{\mathsf{b}} = -H\, D\kappa\, \mathscr{D}^{\circ}(\kappa y)$, $t_q(y) = y\, 1_{|y|>r_q}$.

Proposition. (uniformly in y, N, p)

$$\|b_q^{\mathsf{b}}\|_p \leqslant C\lambda^{7/2} q^{-5/2}, \qquad \|b_q^{\mathsf{I}}\|_p \leqslant C\lambda\, q^{-3/2}(\|t_q\|_p + r_q^{-1}\|y^2 1_{|y|>r_q}\|_p),$$

$$\|b_q^{\mathsf{I}}\|_2^2 \leqslant C\lambda^4 q^{-5}\|t_q\|_6^6.$$

Sources of b^{I} (four groups): the mass term $M' y - J M'\, \kappa y$ ($J = D\kappa$, $M' = \partial_q M$; linear part unclipped, flow computed at κy); the motion of the clip; cubic vertex at y vs correction at κy; chain-rule terms with κ'', κ'''. All vanish if $\|y\|_\infty \leqslant r_q$.

The point

b^{I} is as large as the whole drift, but **supported on large fields**. It is controlled by how often the actual dynamics visits them: $\int q^{-3/2}\lambda \|t_q(Y_q)\|_6^6\, \mathrm{d}q$ (Lecture 7).

The coordinate equation

With $DU = \frac{\lambda}{3} y^3$:

Up to a small Lipschitz perturbation, the forecast solves the cubic elliptic equation

$$K_q \Phi_q = K_q (Y_q + \varrho_q(Y_q)) + \frac{\lambda}{3} Y_q^3, \qquad \varrho_q(y) = C_q (M_q y + \tilde{\ell}_q(y)), \quad \|D\varrho_q\|_{p\to p} \leqslant C (g_q + g_q^2).$$

driven by the field at resolution q.

Summary of today, at every lattice:

- μ_N, M_q and a globally coercive $\bar{\mathscr{F}}_q$, all deterministic;
- the exact FBSDE for (Y, Z);
- a source whose bounded-field part is integrable (two steps), and whose large-field part is controlled by sixth powers of Y on large fields.

Next: estimates on (Y, Z) uniform in N.

The force below the auxiliary scale

The expansion needs g_q small, so $\bar{\mathscr{F}}_q$ is built for $q \geqslant q_*$. Below q_* nothing needs constructing: freeze it,

$$\bar{\mathscr{F}}_q(y) = \bar{\mathscr{F}}_{q_*}(y) + (q_* - q)y.$$

Then $K_q + D\bar{\mathscr{F}}_q = K_{q_*} + D\bar{\mathscr{F}}_{q_*}$: coercivity is automatic, $H_q = H_{q_*}$, and $y_q(x) = y_{q_*}(C_{q_*}K_q x)$.

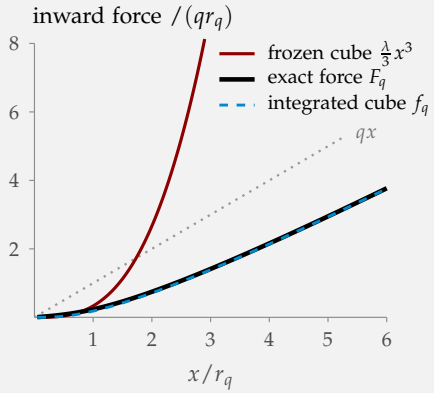
The defect is explicit: $\mathscr{D}_s(y) = -y + \frac{1}{2}\mathsf{C}\mathrm{tr}_{H^2_{q_*}} D^2\bar{\mathscr{F}}_{q_*}(y)$, with $\|\mathscr{D}_s(y)\|_p \leqslant C\|y\|_p$.

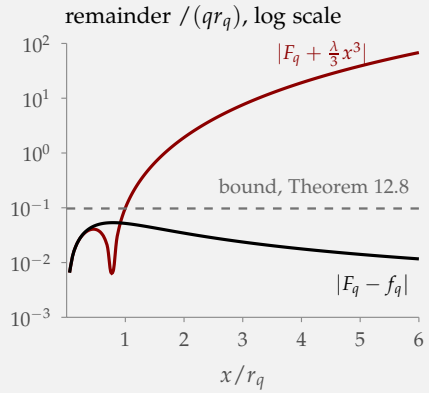
So the system holds on $[q_0, \infty)$ with $\Upsilon_{q_0} = 0$: the initial value is deterministic, with no large mass. q_* stays only in the proofs, where the expansion and the closure of the loop need g_q small.

Why not Polchinski with a frozen cube?

Programme: keep $-\frac{\lambda}{3}\varphi^3$ unclipped, clip the higher orders, pay in the source of $R = F - f$. Test it on one site: $K_q = q$, $S = 1/q$.

$$F_q(x) = -\frac{\lambda}{3}\mathbb{E}_x Y^3 = q(m_q(x) - x), \quad |m_q(x)| \leqslant 2(\lambda/3)^{-1/4} + 2(3q|x|/\lambda)^{1/3}.$$





So $3R/(\lambda x^3) \to 1$: the remainder **cancels the cube**. Field equation: $q\,x = (q - q_0)m_q + \eta_q$, only linear confinement.

The price of clipping

$f^c = f_1 + \mathscr{C} f_2$, with $f_1 = -\frac{\lambda}{3}(x^3 + 3S x)$, f_2 the first tree, $\mathscr{C} f_2 = \kappa_q{}' \cdot f_2 \circ \kappa_q$. For $|x| \geqslant 2r_q$, the Burgers defect is exactly

$$E_q = S^2 f_1 f_1{}' \approx \frac{\lambda^2}{3} S^2 x^5,$$

of order $(q/\lambda)^{1/2}$ at the edge of the clip: not integrable in q.

$$\mathrm{d}R_q = -[E_q + S^2 (f^c)' R_q]\,\mathrm{d}q + \Xi_q^R\,\mathrm{d}W_q, \quad \text{bracket} \approx \frac{\lambda^2}{3} S^2 x^5 - \lambda S^2 x^2 R.$$

The source and the dissipation cancel for the true $R \sim \frac{\lambda}{3} x^3$: separate estimates give nothing.

Tails independent of R: on one site from the posterior moment, but not uniform on the lattice (negative bare mass).

More orders do not help: the tree series has radius $\frac{2}{3} r_q$.

Integrating the cube

Solve the transport part of the flow exactly:

$$\partial_q f + S^2 f f' = 0, \quad f_\infty = -\frac{\lambda}{3} x^3 \quad \Rightarrow \quad f_q(x) = -\frac{\lambda}{3} y^3, \quad q\,y + \frac{\lambda}{3} y^3 = q\,x.$$

$$\sup_x |F_q(x) - f_q(x)| \leqslant 2C_*(\frac{\lambda}{3q})^{1/2}, \quad C_* < \frac{1}{2}.$$

- $y_q(\Phi_q) = \Upsilon_q$: **this is the forecast coordinate**. $F_q - f_q = K_q Z_q$, and Z is the posterior mean minus the mode.
- The defect is $q\,b_q(y)$: bounded for all fields. The trees are in $x \mapsto y$; the cube does not flow in y, so only loops are clipped.
- The cube is coercive in y: $q\,y + \frac{\lambda}{3} y^3 = q\,\Phi_q$.

Not a no-go theorem: it rules out only estimating R as a small perturbation of the frozen cube. Integrating the cube in the original field is possible, but on the lattice its loop defect must still be renormalised; that route is not proved here.